

ALL AROUND THE FARM

LEGUMES ARE VALUABLE.

Furnish Nutritious Feed For Animals and Also Enrich the Soil.
[Prepared by United States department of agriculture.]

Farmers have known for centuries that the clovers, alfalfa, beans and peas enriched the soil, so that wheat, corn, cotton and similar crops gave better yields when following them than when planted after the grains or grasses. The reason for this fact, however, is of more recent discovery than the fact itself.

The soil enriching, leguminous plants possess normally nodules attached to their roots. These are due to the presence of great numbers of bacteria, minute living organisms which enter the roots from the soil. To us their importance lies in the fact that they possess the power of absorbing nitrogen from the air as it circulates in the soil and turning it into compounds which are readily utilized in the growth of the plants. Plants to which these bacteria are not attached derive the nitrogen required for their growth from



Crimson clover, showing the most advanced stage of ripening that is allowable to use for hay.

the soil alone. In consequence leguminous plants in general contain more nitrogen than the nonleguminous. They are therefore richer feeds, and if they are returned in whole or in part to the soil they increase its nitrogen content and thus promote the growth of the following crop. A third advantage in their use is the fact that when fed to live stock the resulting manure has an unusually high fertilizing value.

Under ordinary circumstances the best way to dispose of these crops is to feed them to live stock. All kinds of stock will benefit from them. Alfalfa hay can be substituted in part for such concentrates as wheat bran and cottonseed meal for both dairy and beef cattle. Red, alsike and sweet clover and field peas are less nutritious, but they are nevertheless more valuable than timothy, redtop, orchard grass or other nonlegumes.

Furthermore, the fertilizing value of a leguminous crop need not be lost when it is fed. From 75 to 90 per cent of this fertilizing value can be given to the soil if the manure from the live stock is spread promptly and properly. The small percentage that is lost is insignificant in comparison with the gain to the live stock.

For various reasons, however, some good and some bad, the farmer does not always keep sufficient live stock to furnish enough manure for the requirements of the soil. Under such circumstances a substitute must be found, and the most available one is a leguminous crop, called, when grown for this purpose, a green manure crop. By plowing under this crop humus is added to the soil in much the same way that it is by applications of manure. Because they also add nitrogen legumes are more desirable green manure crops than nonlegumes.

Plowed under crops, however, obviously produce no direct return, and for that reason it is desirable whenever possible that the same land should produce during the season one crop for harvest and one crop for green manure. In the southern part of the northern states grain, early potatoes and a number of other crops are harvested sufficiently early for them to be followed by a catch crop, as a crop following the main one is called.

When there is no time to grow a second crop before cold weather, legumes may still be used as winter cover crops. These serve three purposes—first, to take up available plant food that might otherwise be leached from the soil and lost; second, to protect the land against erosion; third, to supply winter pasture. In the spring they should be turned under for green manure. Red clover, crimson clover and hairy vetch make good cover crops in the north. A common practice is to mix hairy vetch with rye.

Feeding the Cow.

The dairy cow must have a ration from which she can produce milk if you expect her to make a profitable return for the feed.

Just What She Needed

When women complain of weariness, loss of strength and vitality, backache, pains in sides and hips, dull headaches, dizziness, floating specks, and similar ailments, they too often accept these troubles as their lot because they are women, when the ailments may be the results of disordered kidneys. Mrs. Mary V. Bunker, Milton-on-the-Hudson, N. Y., writes: "I am now on my second bottle of Foley Kidney Pills. They seem to be just what I needed." Try them. —Jones Drug Co.

ACT QUICKLY WHEN HOG CHOLERA IS SUSPECTED

Don't take a chance with a sick hog. Act quickly. Get a veterinarian or a trained man immediately. Don't wait a single hour when you suspect disease in your herd. Use the telephone or send to town at once. Only prompt action will stop hog-cholera losses. Every hog saved will help win the war.

This is the message of tremendous importance which the United States Department of Agriculture is carrying to farmers, not only in the great hog belt included in Iowa, Missouri and Illinois and nearby states, which produce 34 per cent of the hog output of the United States, but also to farmers of the south, where in several states in 1916 hog losses due to cholera rose to two or three times the average losses for the country. Hog losses in 1916 by disease rose to 130 per 1,000 in Arkansas, 105 in Louisiana, 100 in Florida, 80 in Georgia, 75 in Tennessee, 70 in Alabama, and 65 in Mississippi, against an average of 48 per 1,000 for the entire country. At least 90 per cent of these losses were attributed to hog cholera.

The time for argument about anti-hog-cholera serum is past, declare the specialists of the Bureau of Animal Industry, United States Department of Agriculture. Explicit figures are readily available which prove unmistakably that, when administered in time by competent persons, hog-cholera serum will keep well hogs from taking the disease for a period of three to six weeks. Not argument, but quick action is imperative to get serum and a trained operator to the farm and treat the hogs. Hogs will be saved and the nation's meat supply increased if every farmer will immediately upon sign of disease call in the nearest available expert to diagnose the complaint and administer whatever treatment is necessary.

The Department of Agriculture is urging that all state, county and local live-stock interests, including state veterinarians, live-stock boards, state college workers, county agents, and local veterinarians, begin immediately to organize anti-hog cholera preparedness campaigns to reach every farmer in every county. Such organization already exists in many places. Where it does not there is need of special and immediate action before the hot-weather conditions favorable to hog cholera come on.

Bankers and other business men can aid in the work of informing farmers where they can get expert help immediately when they suspect cholera in their herds. Lists of accredited persons competent to diagnose cholera and administer serum, together with their addresses and telephone numbers, should be furnished to each farmer in the county, with the appeal that he call for help at once in case of need. Business men may also arrange through the local veterinary authorities or county agents to buy serum at the nearest state serum plant, private plant, or serum depot, and hold it in reserve for emergency use. State and federal animal-industry authorities should be notified at once by telegram of any outbreak.

Because a single outbreak of hog cholera on any farm is a matter of moment to the entire community, and because hog-cholera infection is rapidly spread in many ways, the Department of Agriculture is urging farmers to employ every possible measure to keep the disease out of their herds. The department has issued a special hog-cholera poster—"Stop Hog-Cholera Losses"—which gives specific directions for avoiding infection by sanitary and self-quarantine measures; also directions for the isolation and treatment of cholera-infected hogs, and directions for disinfection and restocking the hog lot. Farmers' Bulletin 834, soon to be published, contains the latest information at the command of the United States Department of Agriculture concerning hog-cholera prevention and the immunization of hogs against the disease.

Does Any Town Want This Couple

A woman writes to the Woman's Home Companion and says in the July issue:

"We are a young couple looking for a town to call our home town, and a position in that town. As we are now situated it is impossible for us to have an interest in the town or to attend church regularly, which is a great trial; and so we have decided, if needs be, to try to secure a position, at less pay, where we could work for our own home, and have a church to attend and be interested in for our little son's sake as well as our own."

"Surely somewhere, in this state, country and town, there is someone looking for an energetic young man, a good steady fellow, who neither gambles, smokes, or so much as tastes a drop of intoxicating beverages, who principally makes his employer's interests his own, and who in turn wishes a position offering a fair livelihood, permanency and a chance for advancement."

"Isn't there some town which would welcome a young couple anxious to lead a useful life and a helpful one to their fellow neighbors?"

Charges Non-Support

Jessie A. Allen, claiming desertion and non-support, has started a divorce suit against Owen Nelson in the circuit court here. There are four children. The couple were married in North Yamhill in April, 1896, and she claims he deserted her May 15, 1907.

Shields Elected Director

Charles Shields was elected director of the Union high school No. 103, West Linn and Willamette, Monday night. W. C. McDonald was re-elected director and Clyde Hughes, clerk, of district No. 34, West Linn.

Audrey Folrich Divorced

Audrey Folrich was granted a divorce from Albert Folrich by Judge J. U. Campbell, of the circuit court, here Saturday. She was given the custody of a minor child.

DAIRY WISDOM.

- Don't sacrifice any better calf from a first class dairy cow.
- Give each calf a chance to drink water at least twice each day.
- Give the cows a good feeding of hay before turning into the fresh pasture.
- Put the cows on the low ground where the coarse grass springs up and grows rankly at the start. If this grass is left uncropped it becomes tough and will not be eaten at all.
- Calves can be raised perfectly on skim milk.

SUMMER SILAGE CROPS.

Provision Should Be Made to Tide Over Scant Pasture Period.

Dairy farmers who have not already made preparations for tiding cows over the period of insufficient pastures in late summer and fall should not forget that many cows went nearly dry last July and August, says a Pennsylvania Agricultural college bulletin. Such conditions may be avoided and incidentally the dairyman's bank account strengthened by a little forethought.

Maintaining the milk flow during the period of insufficient pasture may be accomplished by feeding additional grain, by feeding silage crops and by feeding silage.

Feeding silage crops in addition to the pasture secured by the cows helps to maintain the milk flow at moderate cost. The objection to this method is the labor and planning necessary in planting and cutting such crops at the proper time. Unless one has some silage from last year's feeding silage crops offer the most feasible feeding plan for the coming summer.

The best, cheapest and most practical way to tide over short pastures is to have an ample supply of corn silage. Plan to have enough left over from winter feeding to take care of the short pasture period. This involves less labor than silage crops, and the feed is handy and ready for use whenever needed.

The summer silo should be rather narrow in diameter so that the silage may be fed at the rate of three inches daily to prevent spoiling.

Begin now to plan for a corn acreage sufficient to insure silage for next summer's feeding. It will keep well with the exception of a small amount on top. Put up an extra silo if needed. In the meantime plant some silage crops for use this summer.

CALVES UNPROFITABLE.

Consume More Food in Milk Than They Produce in Meat.

Veal calves from dairy cows generally lose money for their owners, and when these calves are kept until a year old the loss is even greater, is the information brought by cost account records of the Ohio agricultural experiment station for twenty-six herds. The milk used to grow these calves is often of more value as human food than the veal produced.

In the twenty-six dairy herds eighteen owners lost money on raising calves for veal. The average loss on 323 veals was about \$1.95 a head. The average loss on raising 361 calves in thirty herds studied was \$6.90 for the first year. Only five dairymen among the thirty raised calves at a profit.

When dairymen get a high price for milk they would be ahead by selling calves rather than raising such animals for veal or beef. The milk would bring a higher price for human food than the veal produced from it. Pure bred calves usually bring higher prices for breeding purposes, and in such cases it is profitable to raise them.

SWINE ON PASTURE.

Hogs Require Some Grain to Make Economical Gains.

Hogs on pasture require grain for greatest profits in pork production, but a full feed is not economical when pasture is plentiful and grain high priced. When corn alone is fed a limited ration is considerably cheaper because the forage crop takes the place of much of the grain.

Hogs fed all the grain they would eat made more rapid but less economical gains in feeding tests at the Ohio Agricultural Experiment station. They can be marketed sooner, however, which often is an advantage to farmers.

Some protein feed, such as tankage or skim milk, is recommended to be fed in small quantities along with corn and pasture to pigs weighing less than 100 pounds. Pork production is usually too expensive when such feeds are given to large hogs, especially if they are on soy pasture as alfalfa, clover, rape or soy beans, all of which are high in protein.

Care of Horses.

Frequent currying and brushing will insure a healthy condition of the horse. A few minutes spent daily in currying and brushing a horse is time well spent. Aside from giving the animal a clean, glossy appearance, this daily brushing stimulates circulation. The skin is an important excretory organ and must be kept clean and free to do its work.

In the spring the horses should be kept especially clean, as the long hours of work, dust, sweat and heat all combine to make proper care of the skin a necessity. Sore shoulders and other afflictions, due to lack of care on the part of the owner, have often caused great loss of time and money.—Farm and Fireside.

HAVE YOU BEEN SICK?

Then you realize the utter weakness that robs ambition, destroys appetite, and makes work a burden.

To restore that strength and stamina that is so essential, nothing has ever equalled or compared with Scott's Emulsion, because its strength-sustaining nourishment invigorates the blood to distribute energy throughout the body and restores health in a natural, permanent way.

If you are run down, tired, nervous, overworked or lack strength, get Scott's Emulsion to-day. It is free from alcohol. Scott & Bowne, Bloomfield, N. J.

SMALL WHITE MAGGOT IS INJURING CABBAGE PLANTS

Wholesale injury to roots of cabbage and to radishes by a small white maggot is reported daily to the entomology department at O. A. C. Professor Lovett, entomologist at the Oregon experiment station, discusses the pest and its control as follows: "The adult of the cabbage and radish maggot is a small, brown, hump-backed fly, a little more than one-half the size of a house fly. These flies deposit eggs on the soil about the roots of cabbage and radish. These eggs hatch in from twenty to thirty-six hours into a small maggot, which bores its way into the roots of the plants."

"Screening of the permanent radish beds and of the cabbage seed beds to prevent the adult fly getting in to deposit eggs is the most satisfactory method of control. Regular window screen is excellent, or one may use a coarse grade of cheese cloth, running about 16 threads to the inch. The screen should be supported over a frame and the edges wrapped on to the soil so as to make it fly-proof."

"For cabbage and cauliflower plants already set out but not yet infested, there is nothing better than the arched fel discs. These should be cut about two inches across with a slit running to the center. About 25 of them may be placed in a pile and a spike nail driven through the center to form a hole through which the stem of the plant should run. These are placed around the plant and patted down close to the soil. Where the discs cannot be easily obtained, and in case of radishes where it is not possible easily to screen, use tobacco dust. A small amount of this material dusted on the soil around the plant will serve to repel the fly and prevent eggs being deposited. As tobacco dust is quite expensive it may be mixed with an equal amount of wood ashes, air slaked lime, or some similar dust."

"For plants already infested, there is nothing that is really satisfactory, but the following emulsion will kill many eggs and maggots: Crude carbolic acid, 1 pint; fish oil soap, 1 pound; water, 1 gallon. Cut the soap in fine shavings and dissolve in boiling water. Remove from the fire and add slowly the crude carbolic acid, stirring vigorously. Continue the vigorous agitation until an emulsion is formed. To one part of this stock emulsion, add forty parts of water. This weak solution may be poured along the rows of radish plants or about the cabbage plants, using about ½ pint to a cabbage plant. This same material will prevent infection by the maggot if used at eight-day intervals."

DIRECTIONS FOR CANNING FRUIT WITHOUT SUGAR

Fruit for use in pie or salads or as stewed fruit can be put up or canned without the use of any sugar at all, according to the canning specialists of the Department of Agriculture. They, therefore, advise those who, because of the high price of sugar, have been thinking of reducing the amount of fruit they put up, to can as much of their surplus as possible by the use of boiling water when sugar sirup is beyond their means. Any fruit, they say, may be successfully sterilized and retained in the pack by simply adding boiling water instead of the hot sirup. The use of sugar, of course, is desirable in the canning of all kinds of fruits and makes a better and ready-sweetened product. Moreover, most of the fruits when canned in water alone do not retain their natural flavor, texture and color as well as fruit put up in sirup. Fruit canned without sugar to be used for sauces or desserts must be sweetened.

Can the product the same day it is picked. Cull, stem, or seed, and clean the fruit by placing it in a strainer and pouring water over it until it is clean. Pack the product thoroughly in glass jars or tin cans until they are full; use the handle of a tablespoon, wooden ladle, or table knife for packing purposes. Pour over the fruit boiling water from a kettle, place rubbers and caps in position, partially seal if using glass jars, seal completely if using tin cans. Place the containers in a sterilizing vat such as a wash boiler with false bottom, or other receptacle improvised for the purpose. If using a hot water bath outfit, process for 30 minutes; count time after the water has reached the boiling point; the water must cover the highest jar in container. After sterilizing remove packs, seal glass jars, wrap in paper to prevent bleaching and store in a dry, cool place.

If you are canning in tins can it will improve the product to plunge the cans quickly into cold water immediately after sterilization. When using a steam pressure canner instead of the hot water bath, sterilize for 10 minutes with five pounds of steam pressure. Never allow the pressure to go over 10 pounds.

AUTO RIDE APPRECIATED

By Orphans of St. Agnes Baby Home at Parkplace

Last Thursday afternoon eight of the little orphans of the St. Agnes Baby home at Parkplace, accompanied by the sisters, enjoyed a short but delightful ride to Oregon City.

The little ones went to the office of Dr. L. A. Morris, who has kindly donated his services every Tuesday afternoon to take care of the children's teeth.

Barton Barlow took them up in his father's auto and the little ones thought the ride all too short. However, the bright eyes did not fail to recognize some of the kindhearted Oregon City citizens who took them for an outing to Portland about a year and a half ago—one genial friend in particular who promised before long to come after them again.

Miss Alldredge Entertained For the pleasure of Miss Edith All-

ECONOMICAL

No smoke or smell

Roasts Toasts

Bakes Broils

Convenient as GAS

COOK IN A COOL KITCHEN

COOK WITH PEARL OIL

For Sale by
L. Adams Department Store
Hogg Bros.
C. W. Friedrich
Frank Busch
Wilson & Cooke

NEW PERFECTION OIL COOK STOVE Week

dredge, of Oregon City, who is spending several weeks here as a guest of Miss Grace Silcox, the latter entertained informally Thursday night at the Charles A. Johnston home, asking a group of Miss Alldredge's friends here. Dancing afforded amusement, the weather being so pleasant that the merry group danced during the evening on the street pavement in front of the Johnston residence.

In the group were Miss Alldredge, Miss Silcox, Miss Bernice Bright, Miss Edna Baker, Miss Amelia Steeves, Miss Gladys Jane Canter, Donald Silcox, Edward Brown, Chris Wetle, Frank Leshar, George Hostetter, Mr. and Mrs. Thomas Hudson and Mrs. Charles Johnston and Albert Skelton.—The Dalles Chronicle.

Wants Final Papers

Moritz Meyer, native of Switzerland, who took out first papers for citizenship in 1883, eight years after his arrival in America, has made application for final admission to citizenship. He will be examined August 6. Meyer came to Oregon in 1878.

COAL NOTICE!

Diamond and Beaver Hill Coal

from Coos Bay, the cleanest, most suitable House Coal on the market today, can be had for cash at

\$11 per 2001 lbs.

for all ordinary deliveries.

Don't take any chances—make sure of getting your winter's fuel now—there is no time to lose.

E. A. Hackett

Agents for Diamond Coal
Phones: Pac. 247W; Home A22

The Mine Agents, Edlefsen Fuel company, Portland, Oregon, will assist your dealer to secure your fuel supply.

The First National Bank

—of—
OREGON CITY

Capital Stock - \$50,000.00
Surplus - - - \$25,000.00

Member FEDERAL RESERVE BANK
Member OREGON STATE BANKERS ASSOCIATION
Member AMERICAN BANKERS ASSOCIATION

Under Government Supervision

U. S. Depository for Postal Savings Funds
SAFETY DEPOSIT BOXES FOR RENT
INTEREST PAID on Savings Accounts or Time Certificates

We Solicit Your Banking Business

Our Banking Room has been remodeled to meet the requirements of our patrons

Have installed a MODERN and up-to-date VAULT which is mob and burglar proof.

OFFICERS—

D. C. LATOURETTE, President
M. D. LATOURETTE, Vice-Pres.
F. J. MEYER, Cashier



(Established 1889)

OPEN 9 a. m. to 3 p. m.